ETH zürich

Mobile Computing

Socio-Economic and Humanitarian Use

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems - Specialized Applications
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

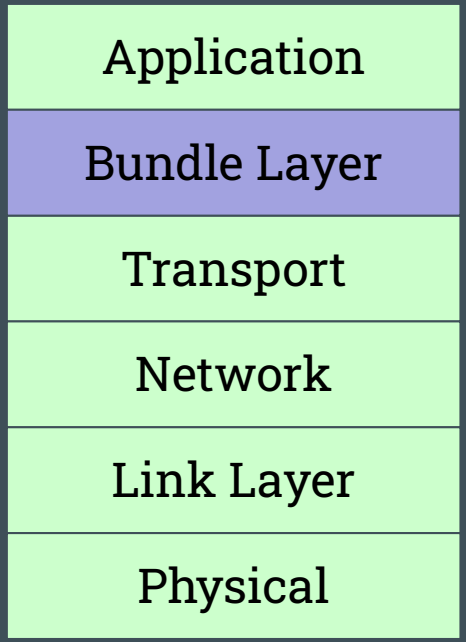
Tournament

Delay Tolerant Networking, DTN

- The DTN idea originated from the Interplanetary Networking Special Interest Group (IPNSIG)
  - Deep space networks and spacecraft communications
- Other use cases:
  - Networks with intermittent connectivity (remote areas)
  - Military networks
- Can tolerate delay or disruption, or disconnection
- Data travels along with moving devices

DTN Layer in the Protocol Stack

- Creates time-stamped (data units)
  - Open for different DTN routing schemes
  - Supports TCP and UDP convergence layer
- IETF Standards
  - rfc 4838: Architecture
  - rfc 5050: Bundle Protocol Specification
- Challenges
  - Unpredictable link error rates
  - Unpredictable network capacity: Low with a small number of devices, but high with a large number of devices



Case 1: Project with ICRC ...

Gaza Strip

People • Humanitarian Aid • Critical Infrastructure

- Hospital
- Field Hospital (humanitarian)
- Humanitarian Aid Location
- City / Town
- Port
- Airport (non-operational)
- Water Desalination Plant
- Power Plant (non-operational)
- Key Infrastructure
- Main Road
- Other Road
- Boundary (Gaza Strip)
- Regional Boundary
- Built-up Area

N

0 5 10 km

MEDITERRANEAN SEA

Erez Crossing (currently closed)

Beit Lahia

Beit Hanoun

Jabalia

Gaza Port (limited operation)

Gaza City

Al Karama Field Hospital

Nuseirat

Bureij

Middle Area Desalination Plant

Deir al-Balah

Humanitarian Aid Hub (Deir al-Balah)

ISRAEL

Khan Younis

Rafah

Yasser Arafat International Airport (non-operational)

Rafah Crossing (partially open)

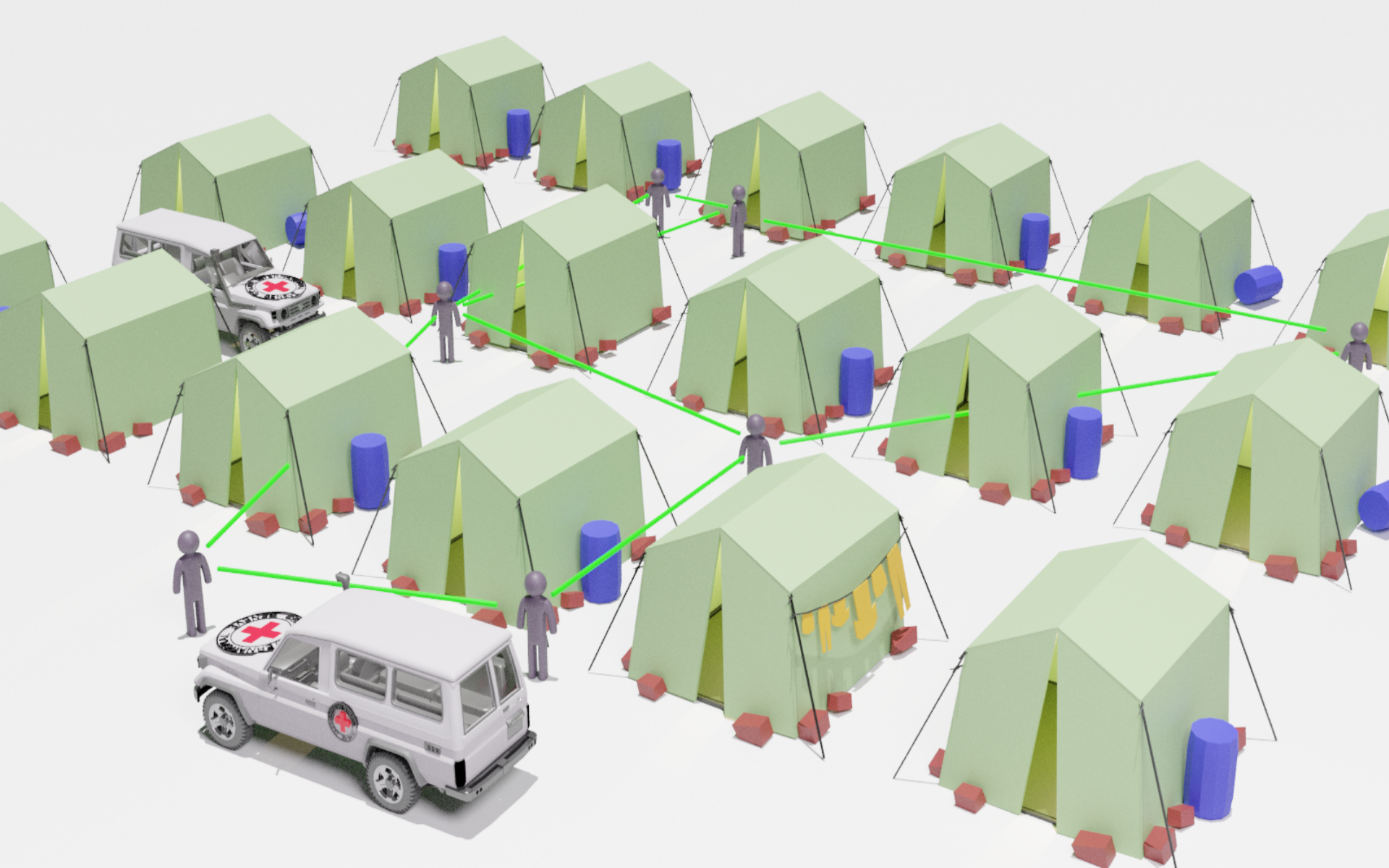
To Egypt

EGYPT

Kerem Shalom

Humanitarian Aid Location (Deir al-Balah)

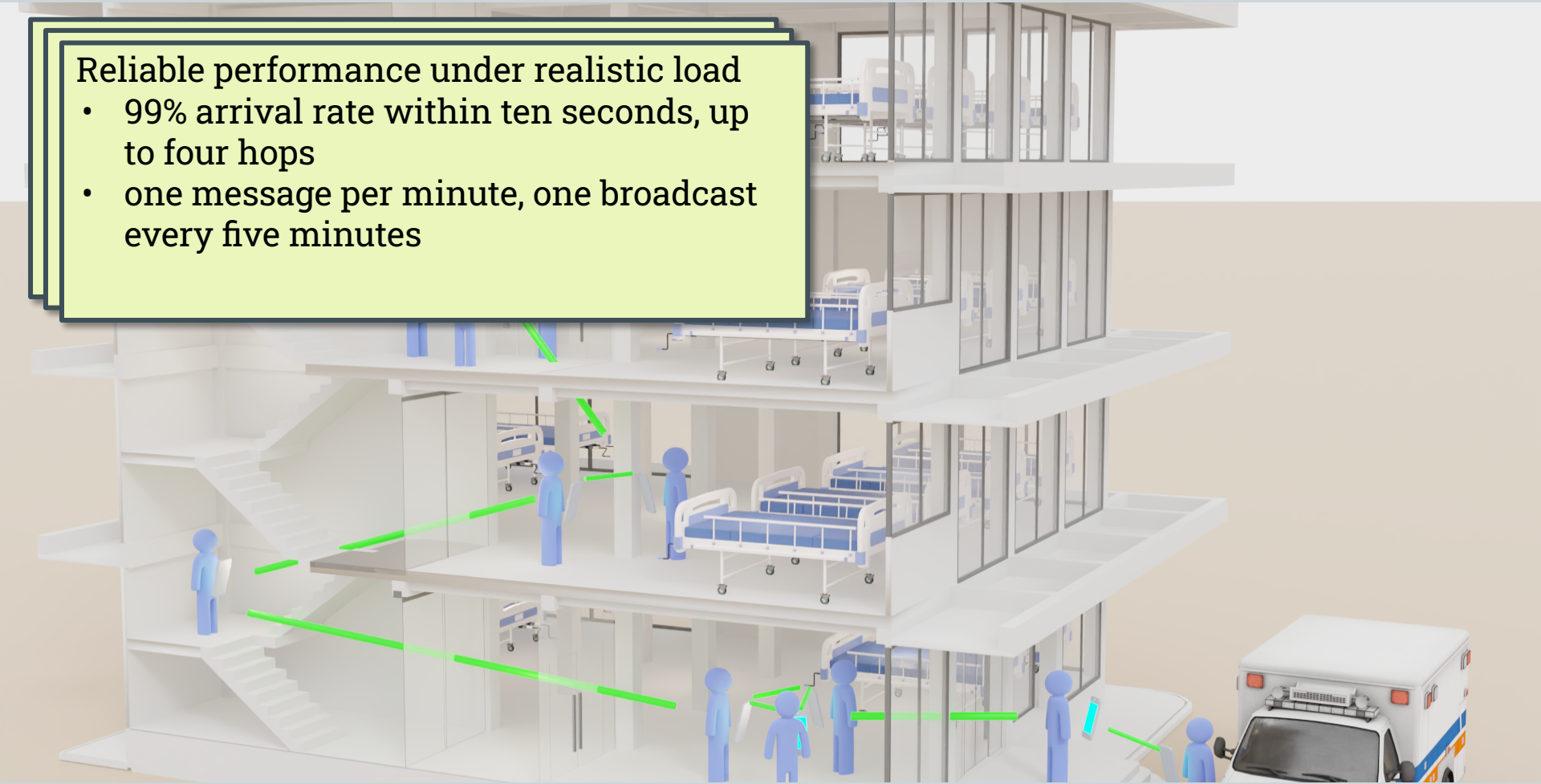
Food · Water · Medical Care · Shelter



Wireless Mesh Networking

- Goal: A mobile app to create a network of mobile devices, operating without any additional telecom infrastructure
- Improve humanitarian workers communication during emergencies
- Adhere to ICRC’s security and privacy constraints
- General relevance for the humanitarian sector in disaster relief (earthquakes, power outages)
- Duration: Sep 2024 … Feb 2026

Wireless Mesh Networking

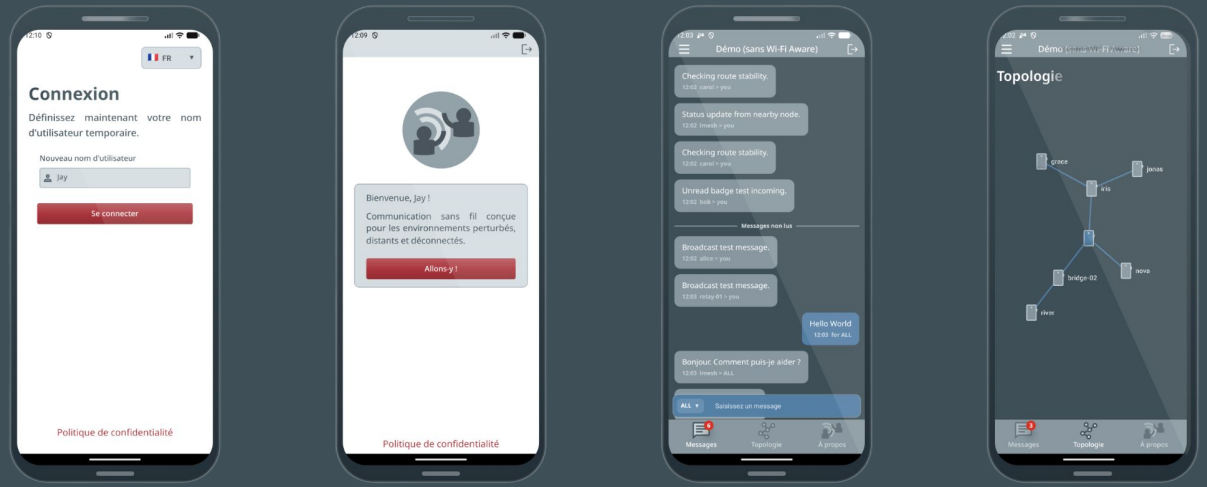


Reliable performance under realistic load

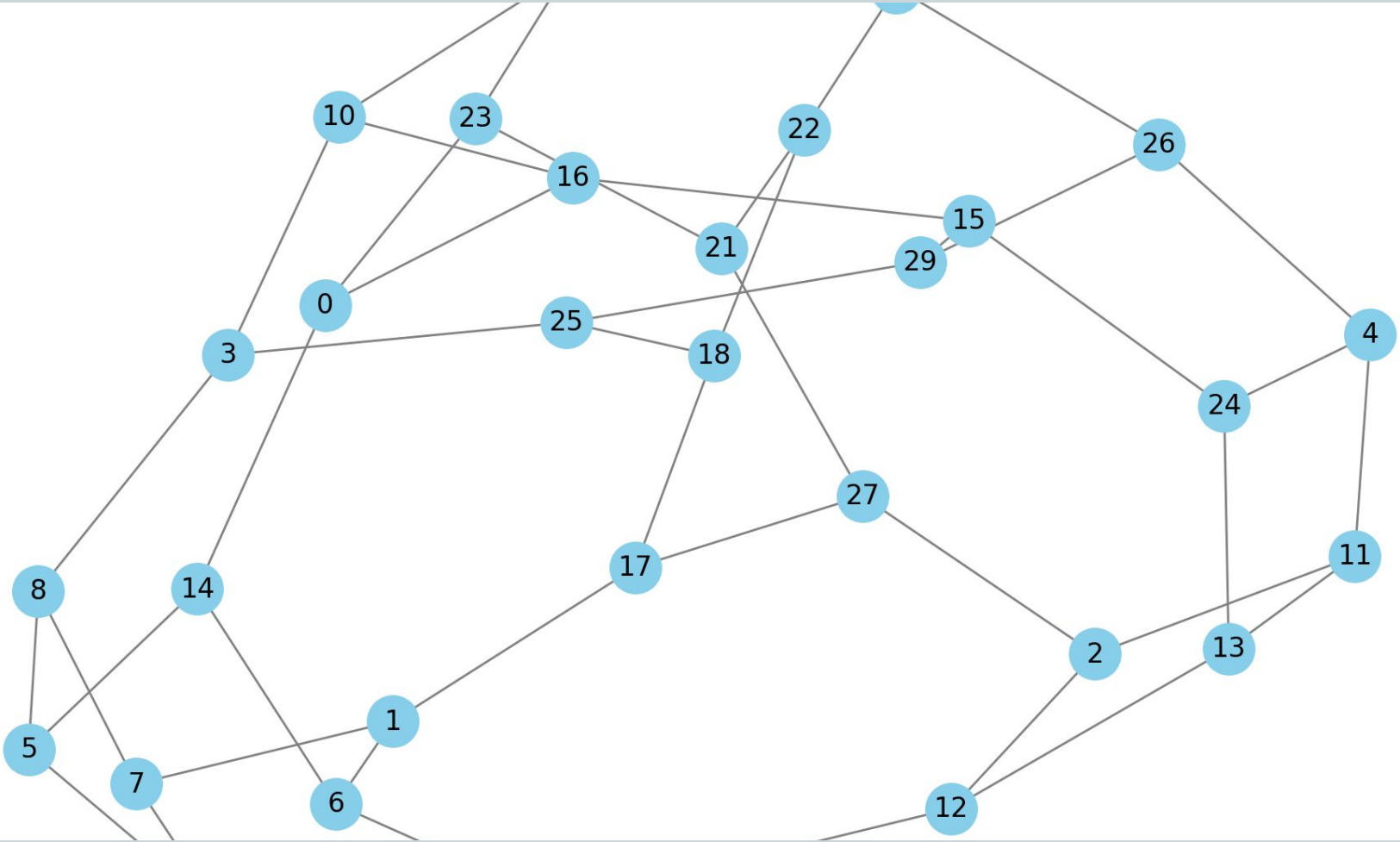
- 99% arrival rate within ten seconds, up to four hops
- one message per minute, one broadcast every five minutes

Prototype

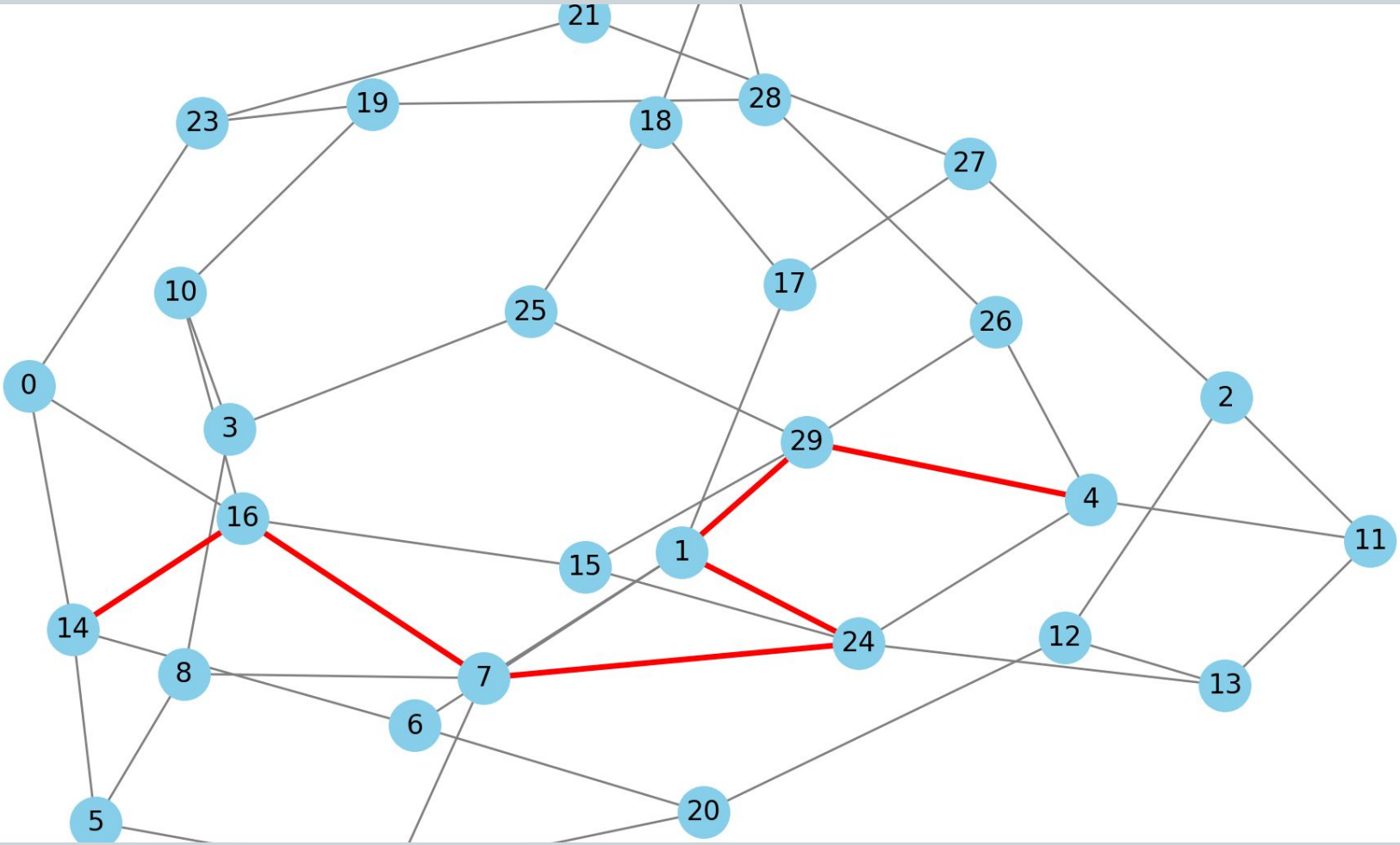
- Experimental wireless communication when cellular and satellite networks are unavailable
- Messaging, local topology visualization, and simple onboarding for disrupted environments



Layer 2: Wi-Fi Aware Mesh Networking

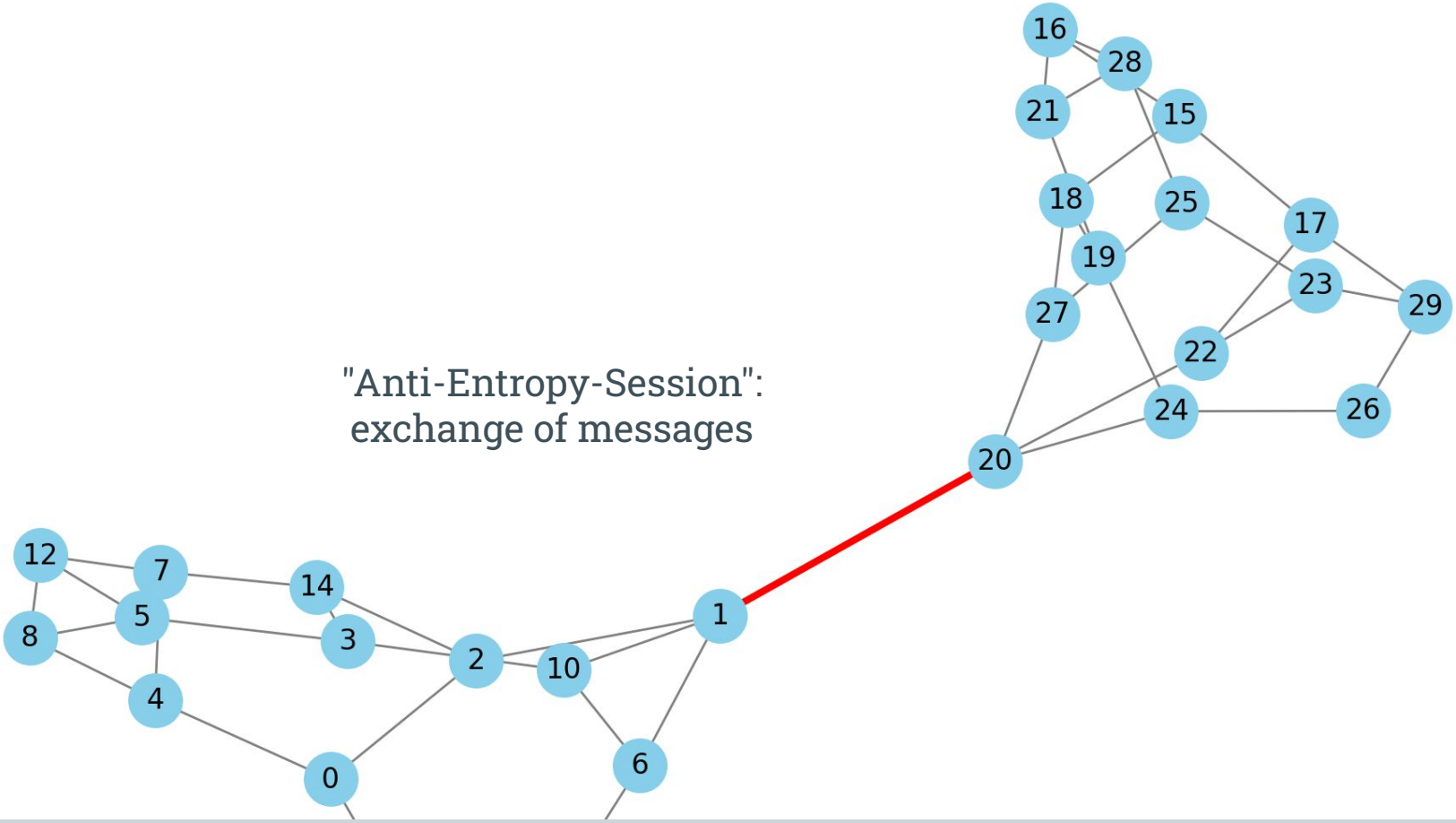


Layer 3: Routing



Layer 4: Delay Tolerant Networking

"Anti-Entropy-Session":
exchange of messages



What is the Wi-Fi Aware Protocol

What is the Wi-Fi Aware Protocol

- Allows nearby devices to discover and communicate with each other — without requiring an internet connection or a traditional Wi-Fi access point
- Original use cases: file sharing, multiplayer gaming, smart home, location-based services
- Devices can detect and connect when in physical proximity
- Designed to be energy-efficient, making it suitable for mobile and IoT devices
- Apps discover nearby devices even when not actively running (Android)
- Already supported on Android, just recently iOS, too!

Case 2: Mobile Empowerment, South Africa

Pretoria – Moloto Corridor

Key Locations: Pretoria, Hammanskraal, Moloto, Vlaklaagte, Kwaggafontein

Connecting communities in Gauteng and Mpumalanga

Route Summary (via R101 and R573)

Pretoria → Hammanskraal	Hammanskraal → Moloto	Moloto → Vlaklaagte	Vlaklaagte → Kwaggafontein
± 40 km (R101)	± 37 km (R573)	± 45 km (R573)	± 10 km (R573)

URBAN CITY
WITH CELLULAR
COVERAGE
Pretoria
Hammanskraal
school

ENTER | EXIT
PUTCO